

Problem 25.21

Payments of \$3650 per year are made continuously over a five-year period. Find the present value of this continuous annuity two years prior to the first payment. Given the nominal rate $i^{(365)} = 8\%$.

Solution

We are given the nominal interest rate convertible daily:

$$i^{(365)} = 0.08.$$

The corresponding annual effective interest rate is

$$i = \left(1 + \frac{0.08}{365}\right)^{365} - 1.$$

Thus,

$$i = 0.08327757.$$

The corresponding force of interest is

$$\delta = \ln(1 + i).$$

So

$$\delta = \ln(1.08327757) = 0.07999123.$$

Also,

$$v = \frac{1}{1 + i} = 0.92312443.$$

The value at the time of the first payment of the continuous annuity is

$$3650\bar{a}_{\overline{5}|}.$$

Since the present value is required two years prior to the first payment, we multiply by v^2 :

$$PV = 3650v^2\bar{a}_{\overline{5}|}.$$

For a continuous annuity,

$$\bar{a}_{\overline{5}|} = \frac{1 - e^{-5\delta}}{\delta}.$$

Therefore,

$$PV = 3650(0.92312443)^2 \left(\frac{1 - e^{-5(0.07999123)}}{0.07999123} \right).$$

Thus,

$$PV = 12818.13359.$$

Therefore, the present value is

$$\boxed{12818.13}.$$